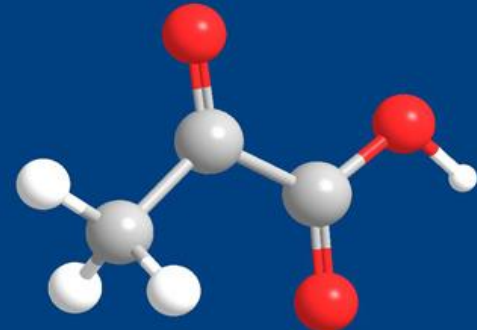
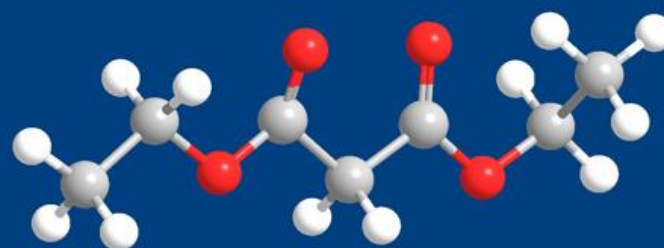




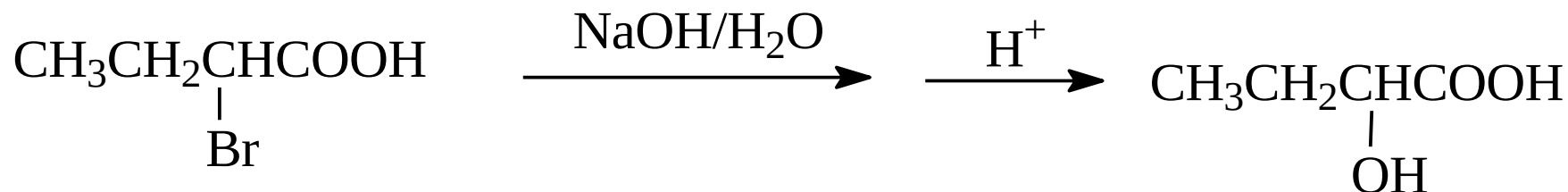
第十三章 取代酸 和 β -二羰基化合物

Substituted Carboxylic Acids and
 β -Dicarbonyl Compounds

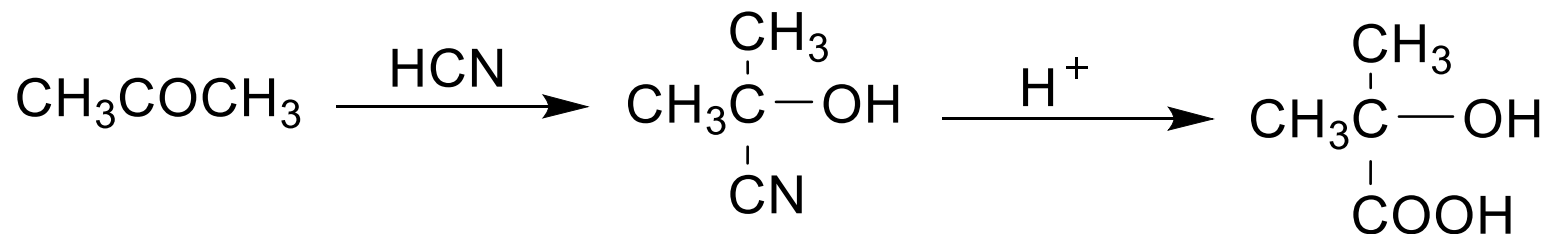


羧酸的制备

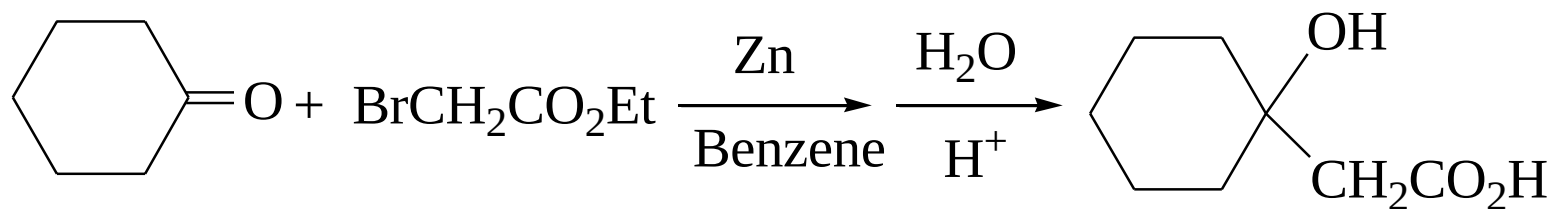
- 卤代酸水解



- 羟基腈水解

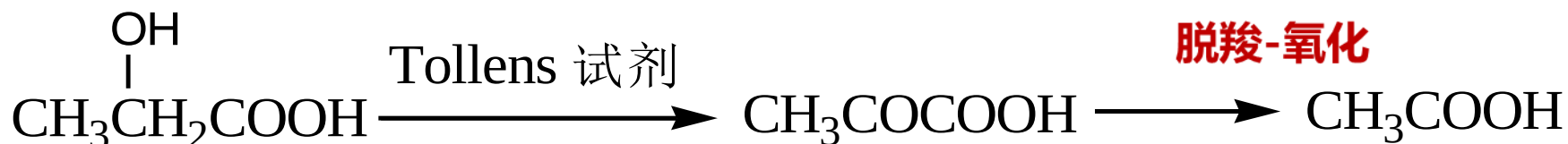


- Reformatsky反应



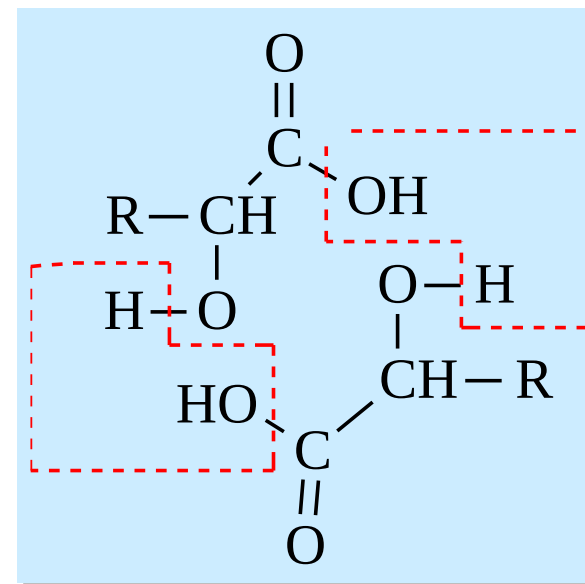
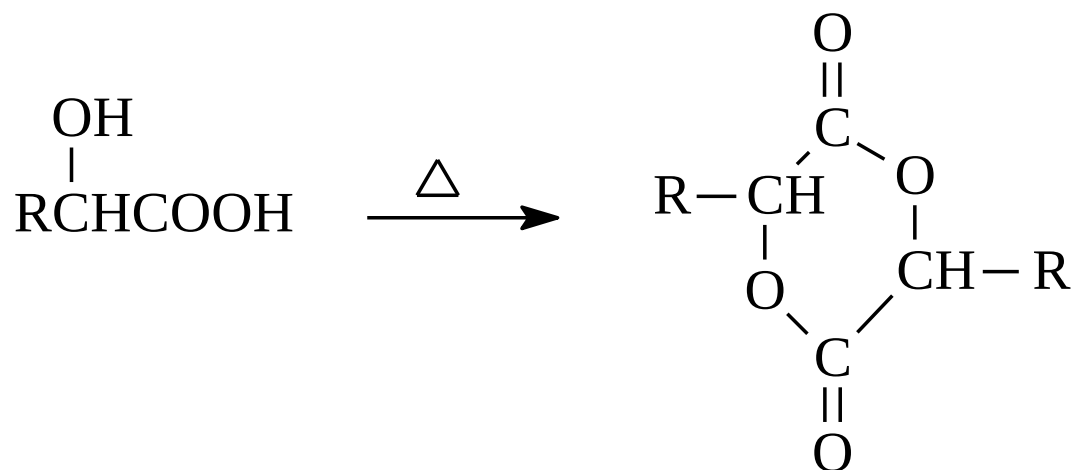
羧酸的化学反应

- 氧化反应

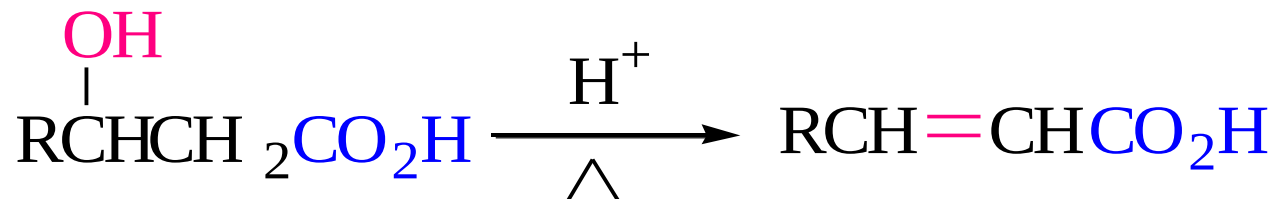


- 脱水反应

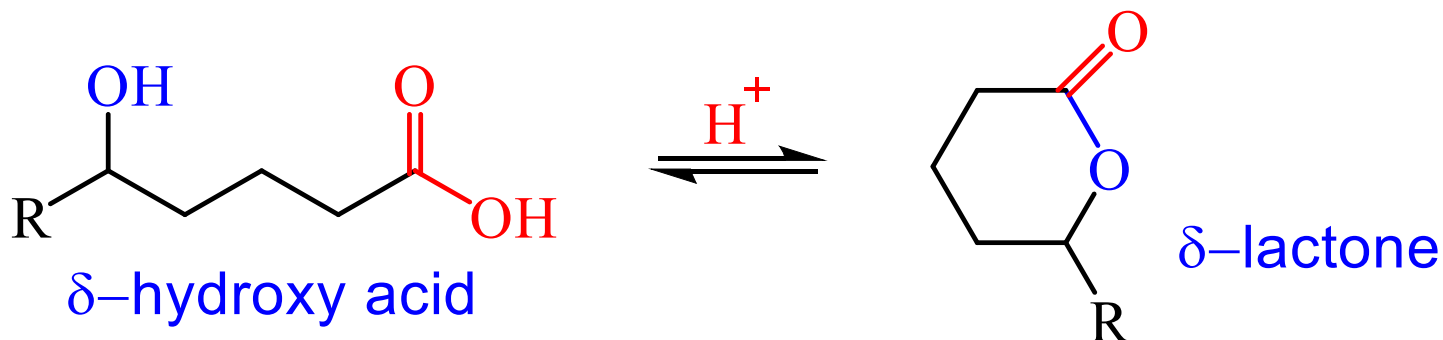
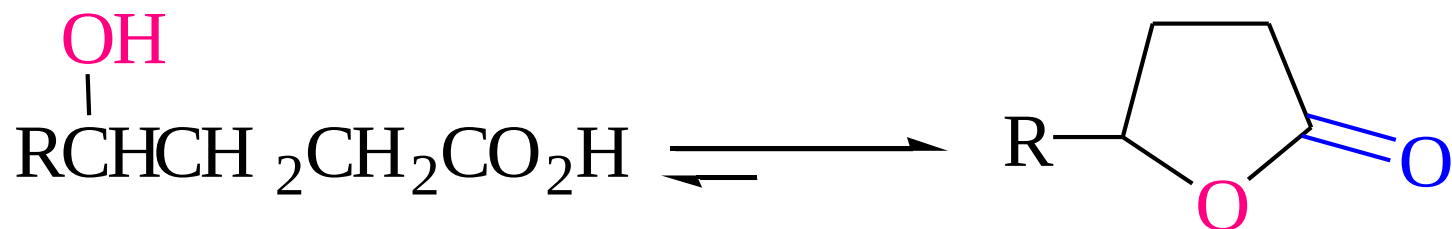
(1) α -羟基酸



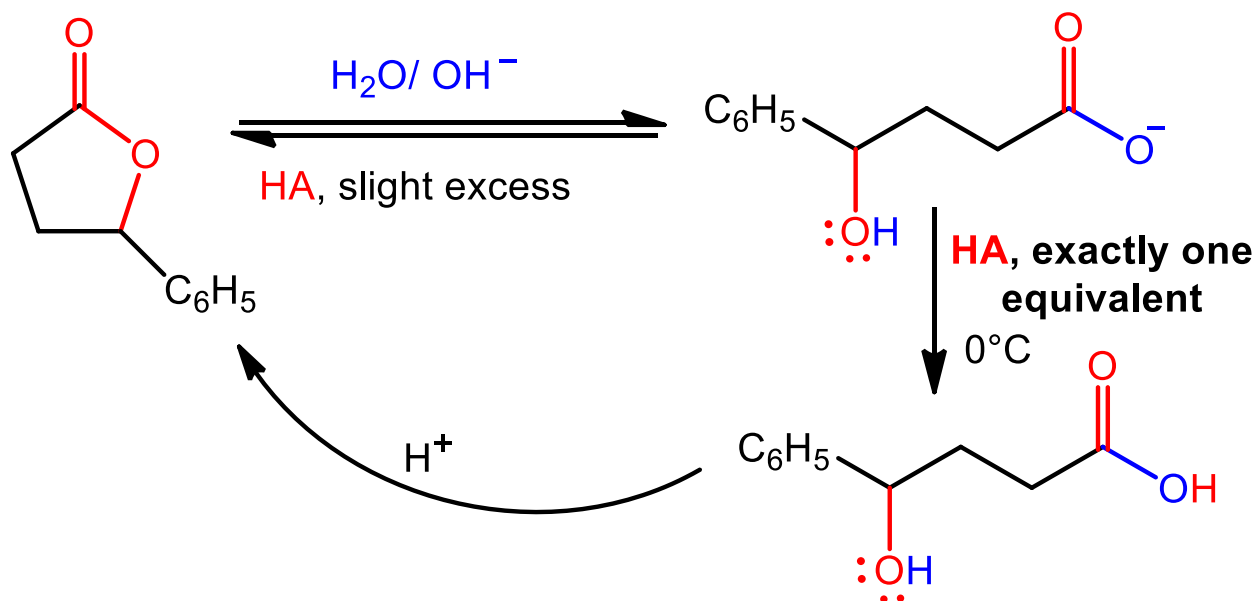
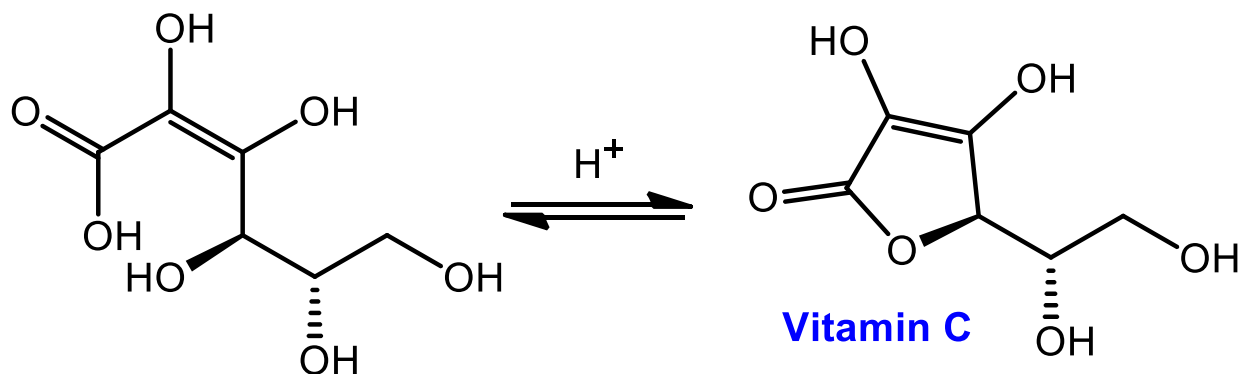
(2) β -羟基酸



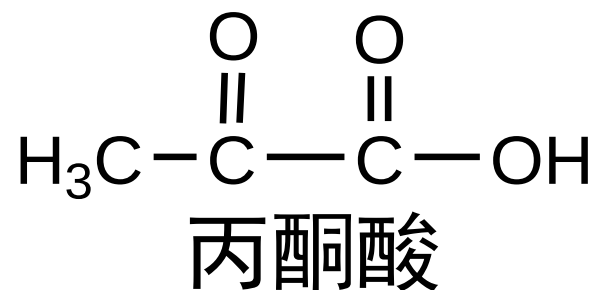
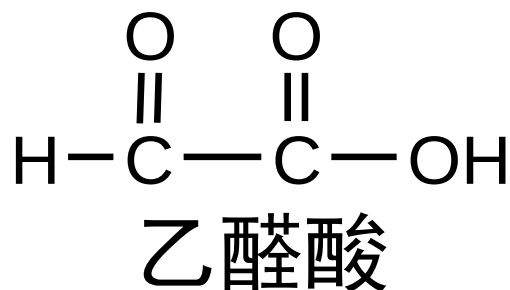
(3) γ -羟基酸



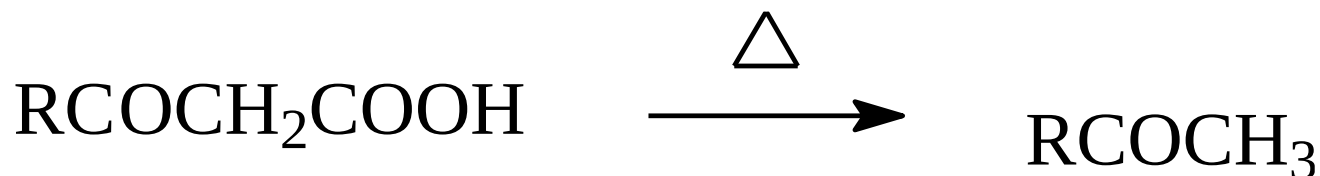
Lactones



α -二羰基化合物



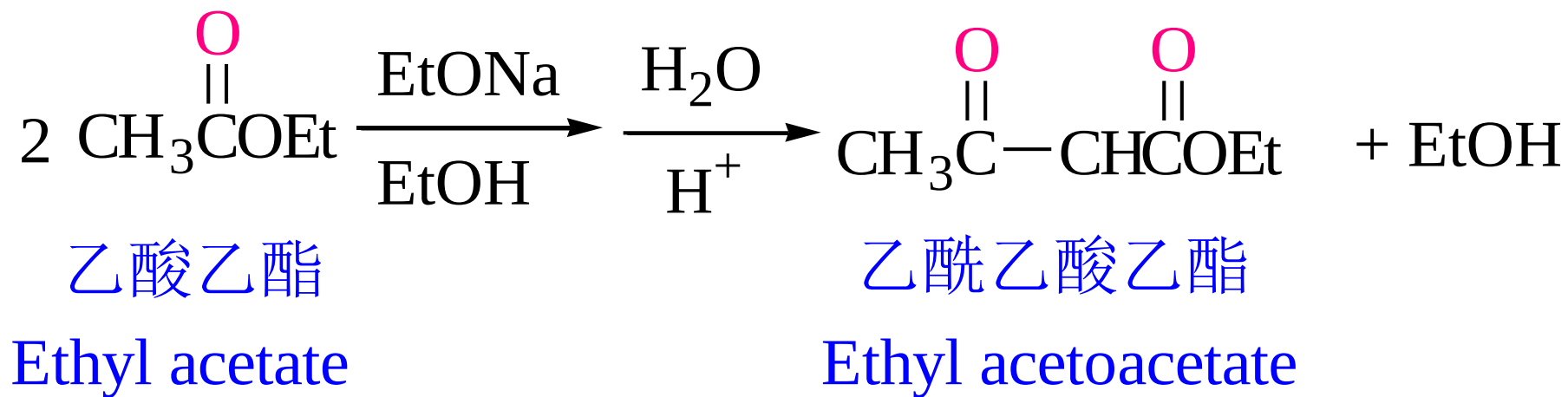
- 酸性大于羧酸
- α 或 β -酮酸受热易脱羧



β -二羰基化合物

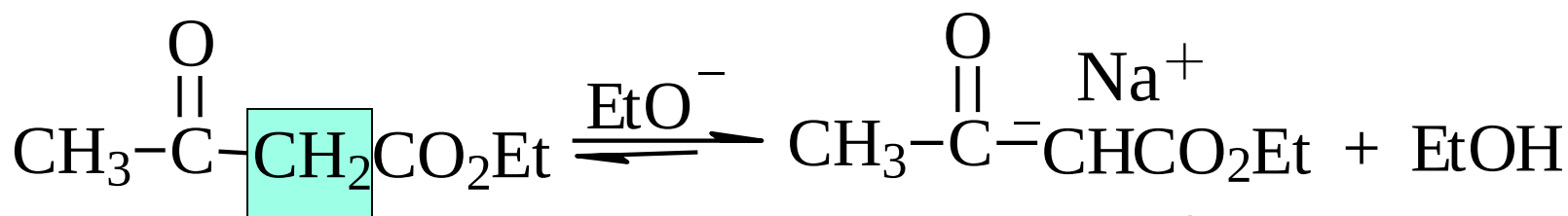
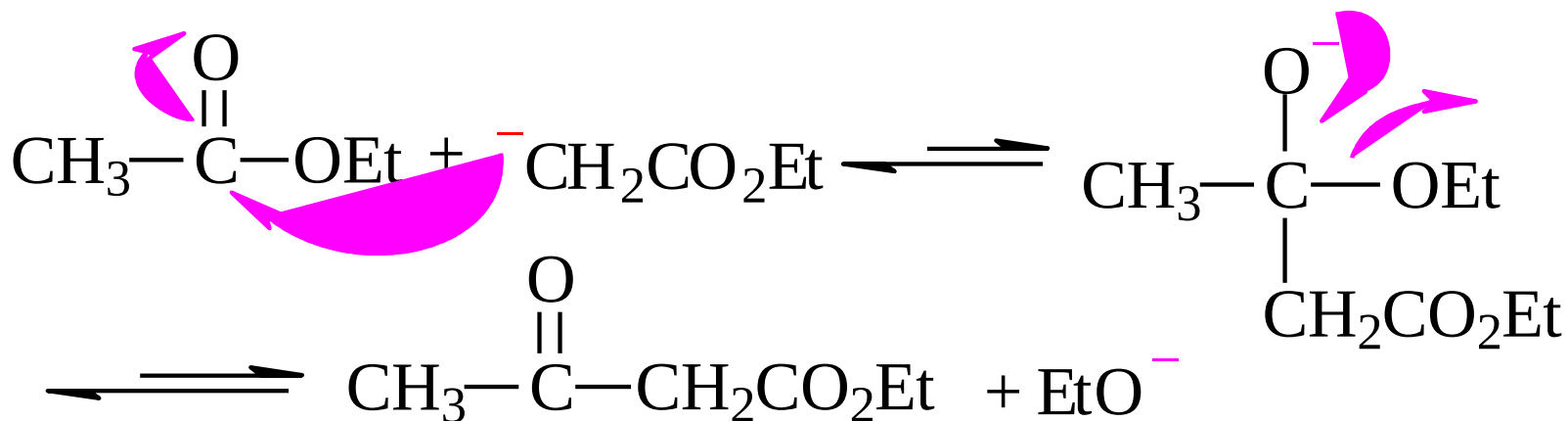
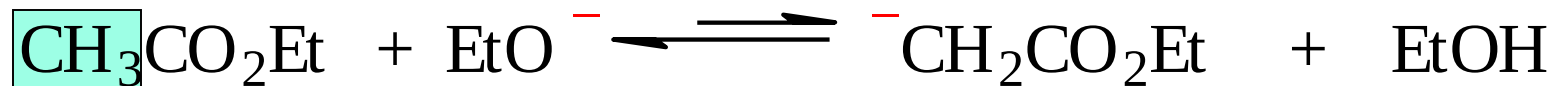
1. β -酮酸酯的形成 (机理见12章)

Claisen 酯缩合

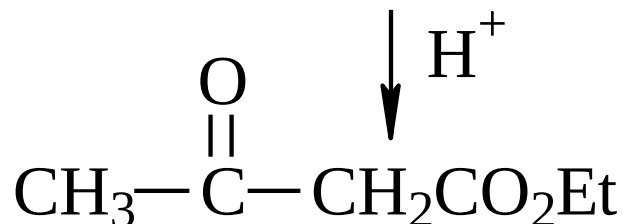


反应机理

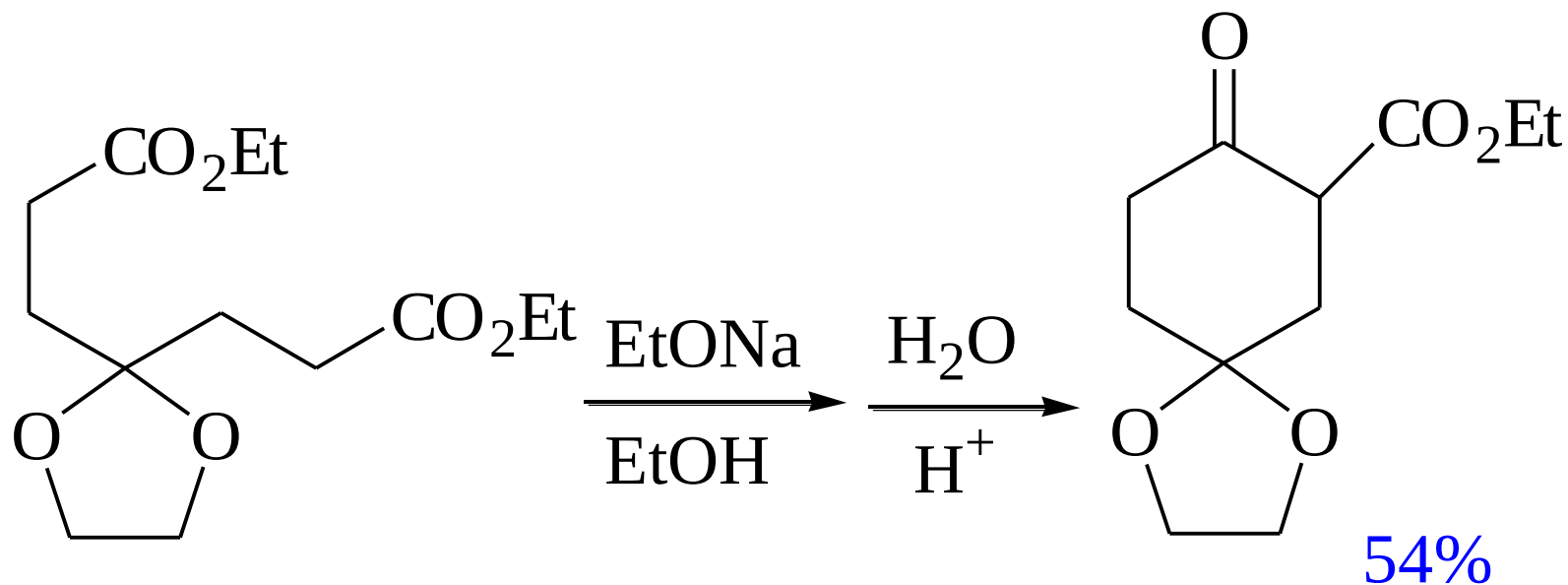
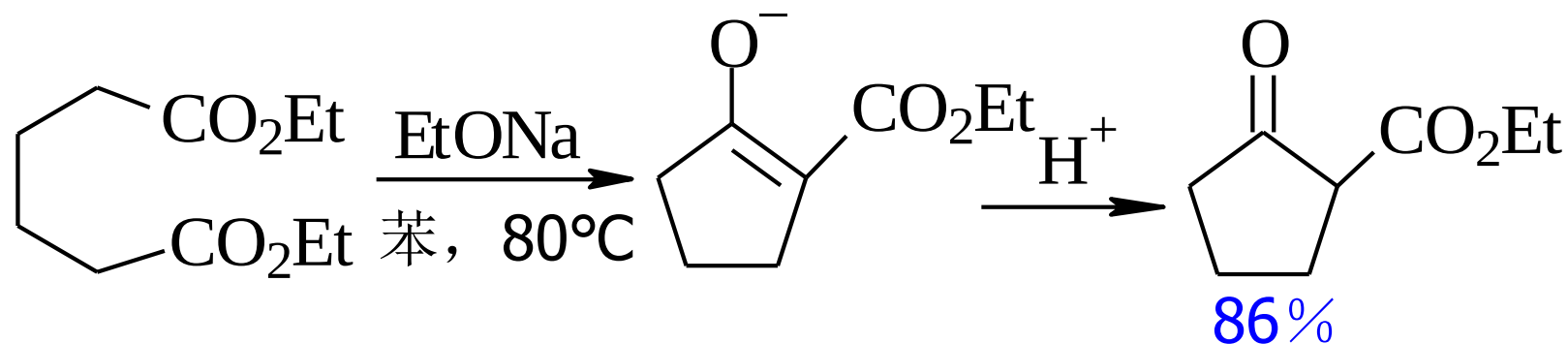
pKa 24.5



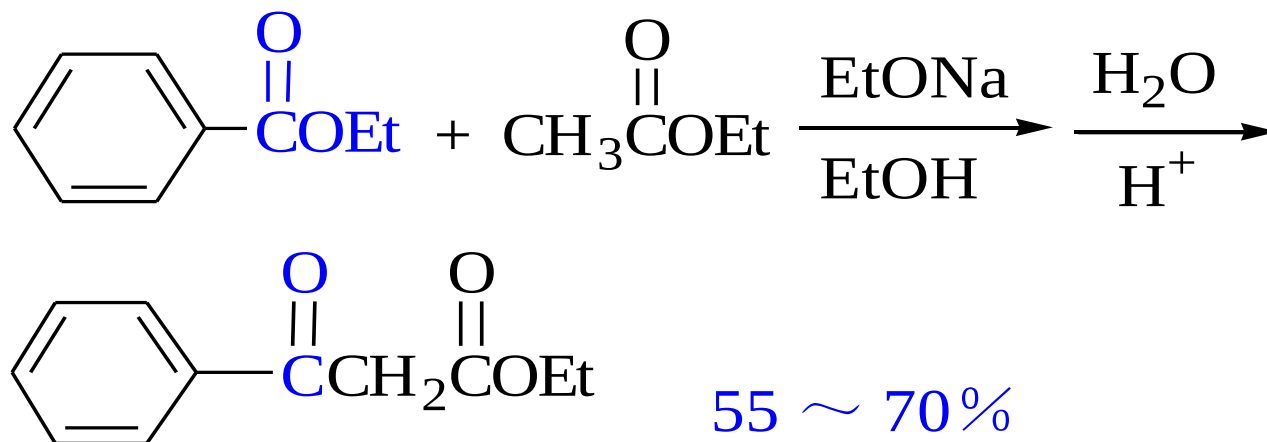
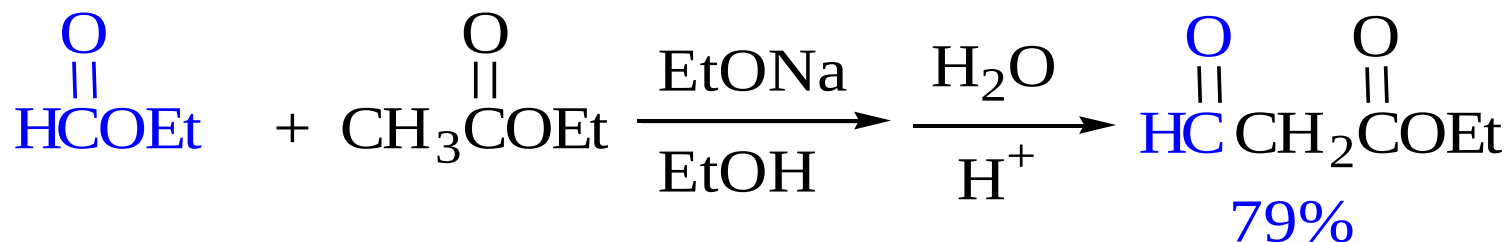
pKa ~11



Dieckmann反应

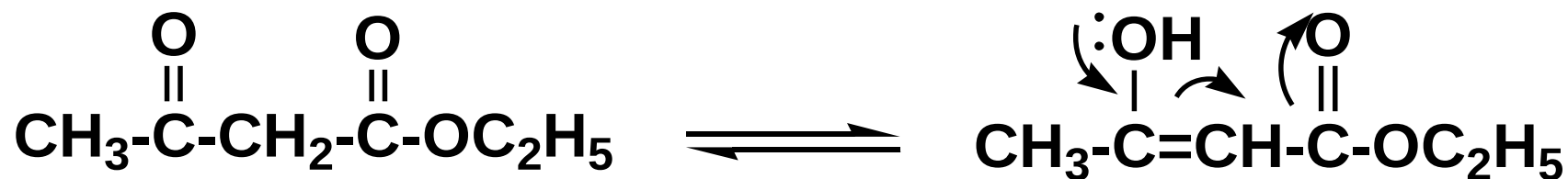


不对称二酯和混合酯的缩合

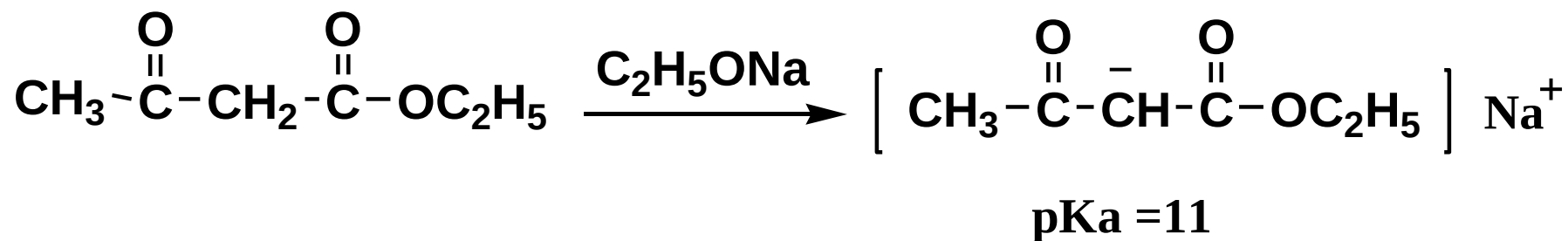


β -二羰基化合物

2. β -酮酸酯的互变异构现象



亚甲基活泼氢的性质：酸性

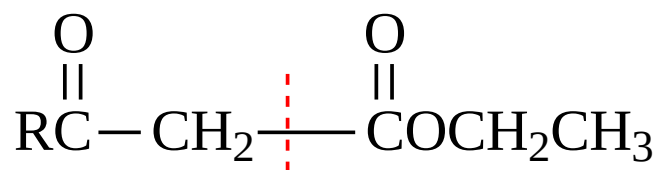


Keto and Enol Tautomers

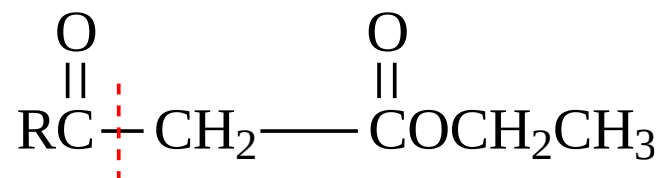
酮式和烯醇式虽然共存于一个平衡体系中，但在绝大多数情况下，酮式是主要的存在形式。但是随着 α 氢的活泼性增大，失去氢后形成的碳负离子的稳定性增大，烯醇式也可能成为平衡体系中的主要存在形式：

化合物	酮式和烯醇式的互变异构	烯醇式含量 (%)
丙酮	$\text{CH}_3\overset{\text{O}}{\parallel}\text{CCH}_3 \rightleftharpoons \text{CH}_3\overset{\text{OH}}{\mid}\text{C}=\text{CH}_2$	0.00015
丙二酸二乙酯	$\text{C}_2\text{H}_5\text{OC}\overset{\text{O}}{\parallel}\text{CH}_2\text{OC}_2\text{H}_5 \rightleftharpoons \text{C}_2\text{H}_5\text{OC}\overset{\text{OH}}{\mid}=\text{CHC}\overset{\text{O}}{\parallel}\text{OC}_2\text{H}_5$	0.0077
乙酰乙酸乙酯	$\text{CH}_3\overset{\text{O}}{\parallel}\text{CCH}_2\overset{\text{O}}{\parallel}\text{COC}_2\text{H}_5 \rightleftharpoons \text{CH}_3\overset{\text{OH}}{\mid}\text{C}=\text{CHC}\overset{\text{O}}{\parallel}\text{OC}_2\text{H}_5$	7.3
2, 4-戊二酮	$\text{CH}_3\overset{\text{O}}{\parallel}\text{CCH}_2\overset{\text{O}}{\parallel}\text{CCH}_3 \rightleftharpoons \text{CH}_3\overset{\text{OH}}{\mid}\text{C}=\text{CHC}\overset{\text{O}}{\parallel}\text{CH}_3$	76.5
1, 1, 1-三氟-2, 4-戊二酮	$\text{CH}_3\overset{\text{O}}{\parallel}\text{CCH}_2\overset{\text{O}}{\parallel}\text{CCF}_3 \rightleftharpoons \text{CH}_3\overset{\text{O}}{\parallel}\text{CCH}=\overset{\text{OH}}{\mid}\text{CCF}_3$	99

β -酮酸酯的水解



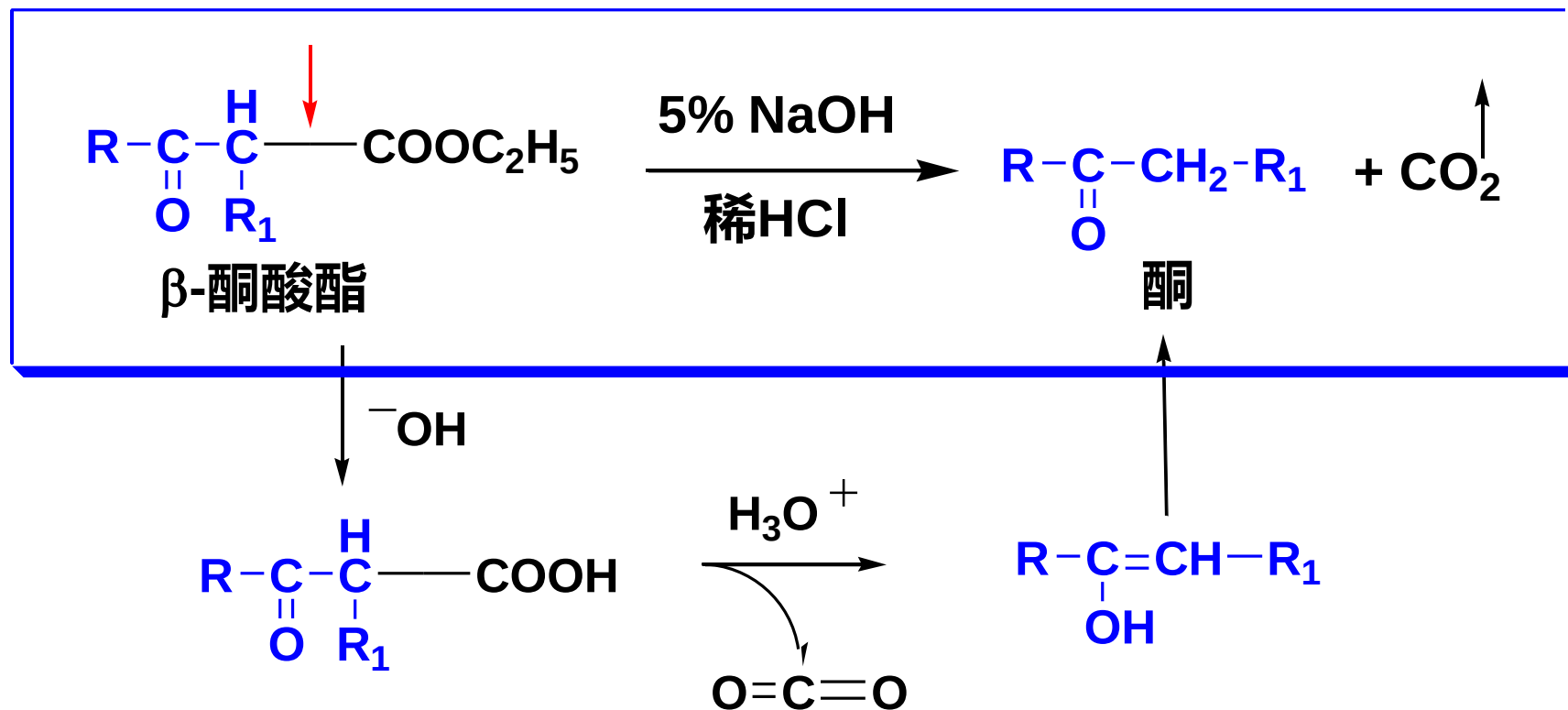
酮式水解 **掌握**



酸式水解 **了解**

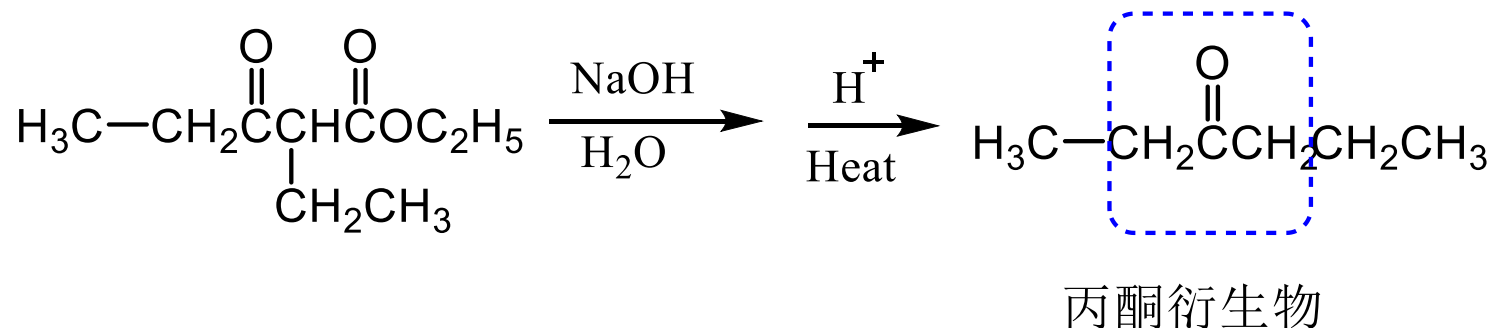
β-酮酸酯的水解

酮式分解：“三乙”在稀碱中水解，酸化后加热脱羧得丙酮；或者“三乙”在酸中加热脱羧得丙酮

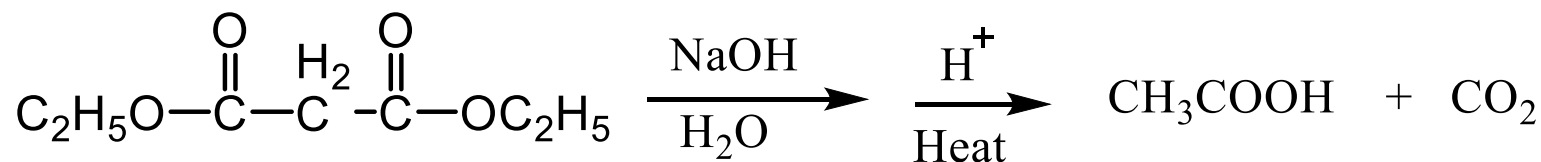


β-酮酸酯的水解

酮式分解: β-羰基乙酸乙酯在稀碱中水解除羧基转化为酮的反应:

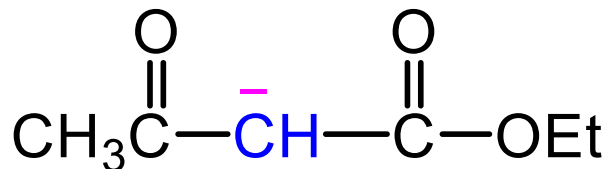


丙二酸二乙酯经酮式分解得乙酸或乙酸的衍生物:

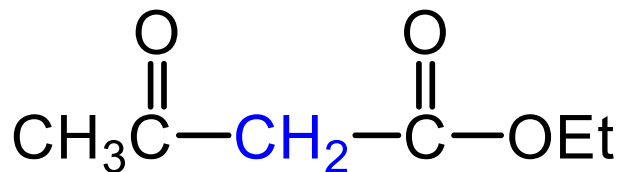
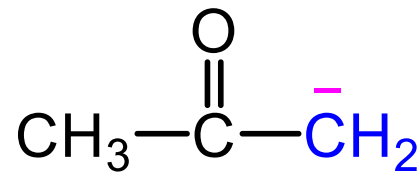


三乙以及酮式分解的应用

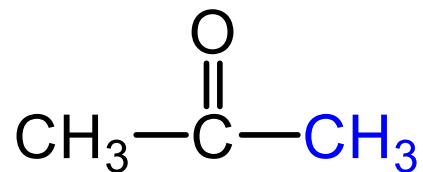
合成烷基酮:



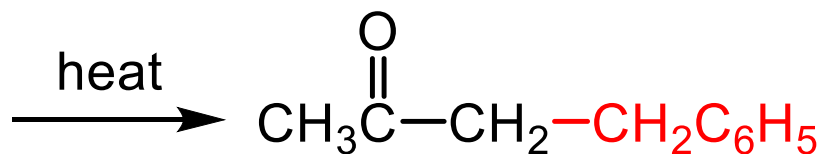
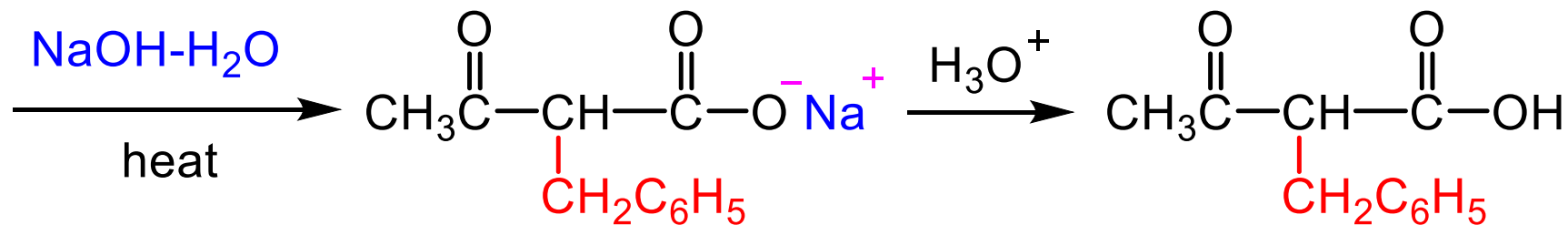
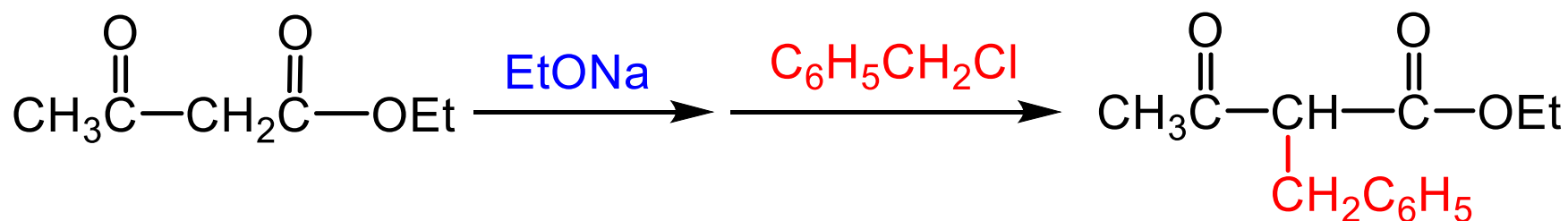
is the synthetic equivalent of

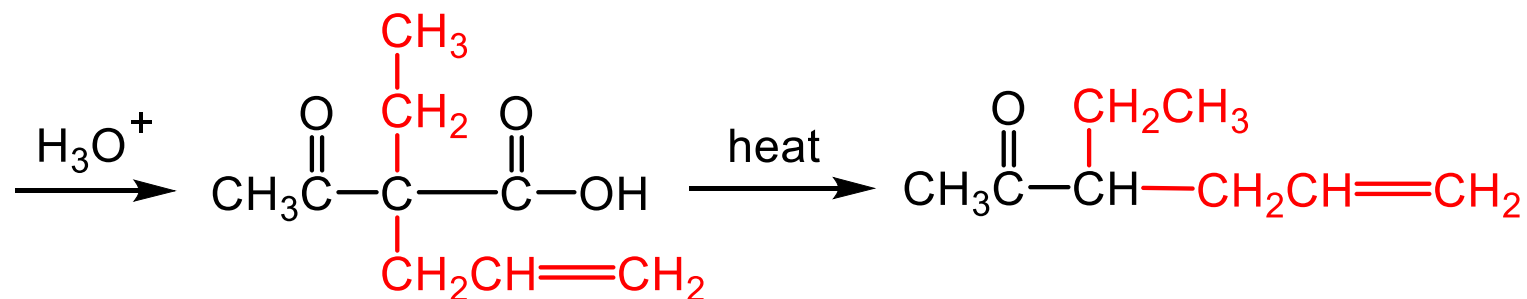
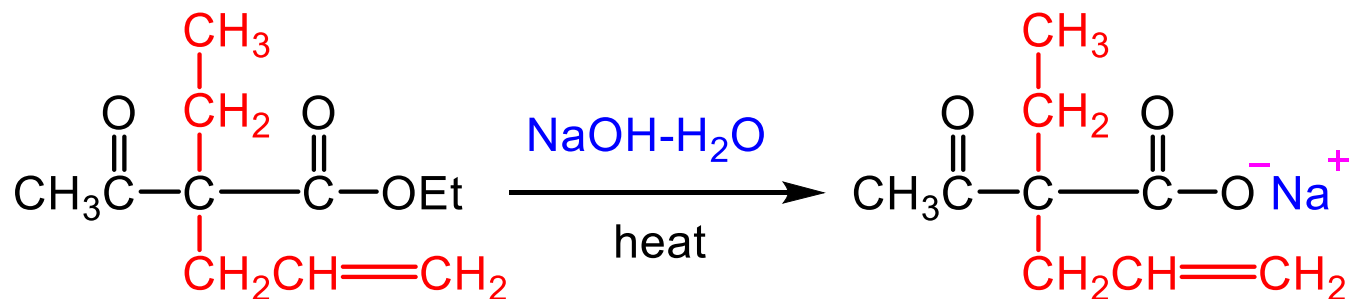
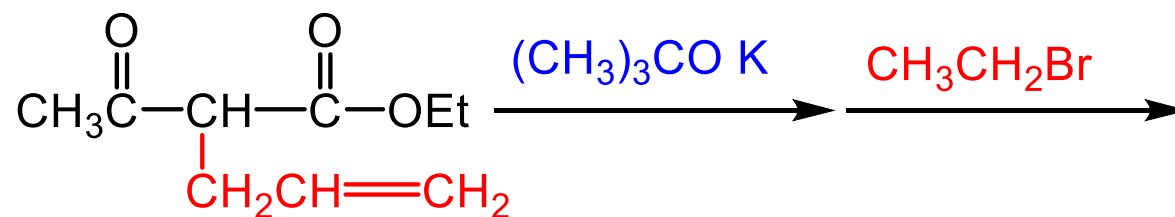
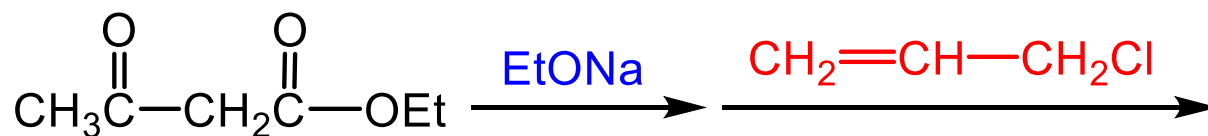


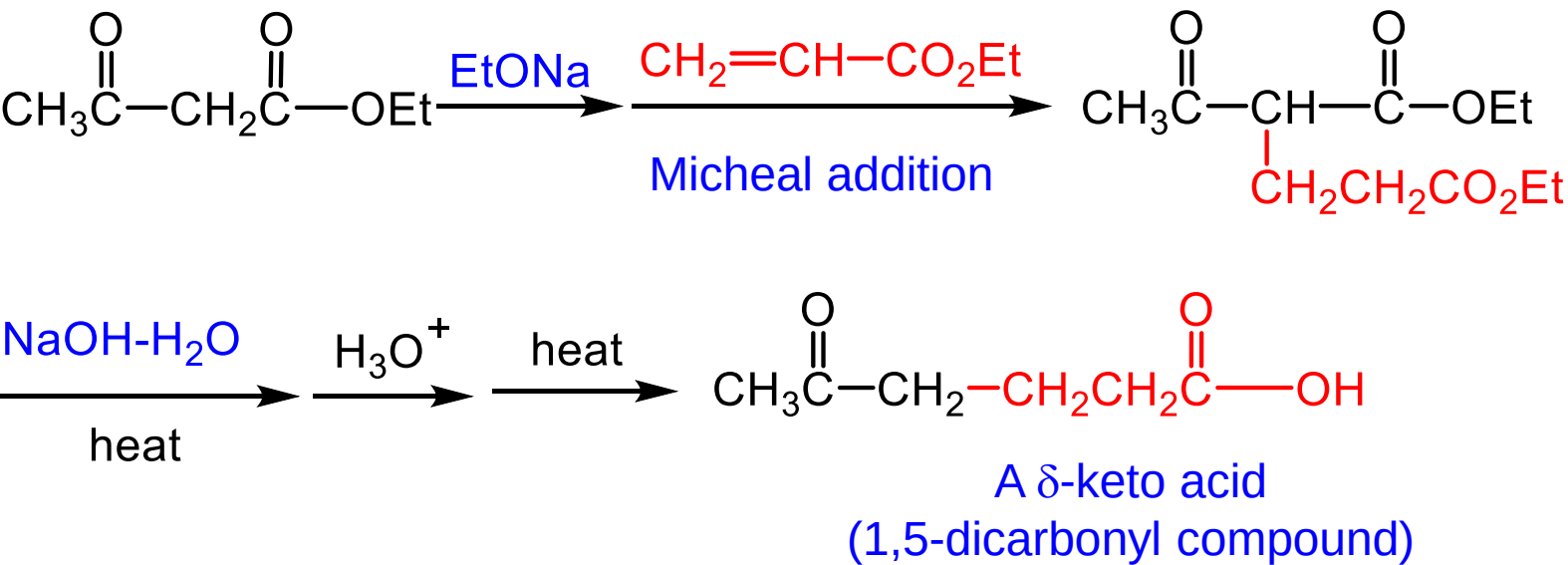
pK_a 9~11



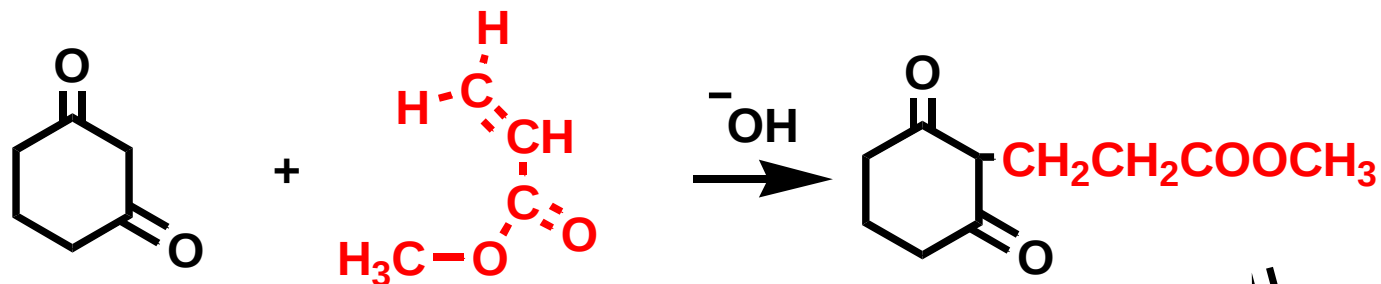
pK_a 19~20



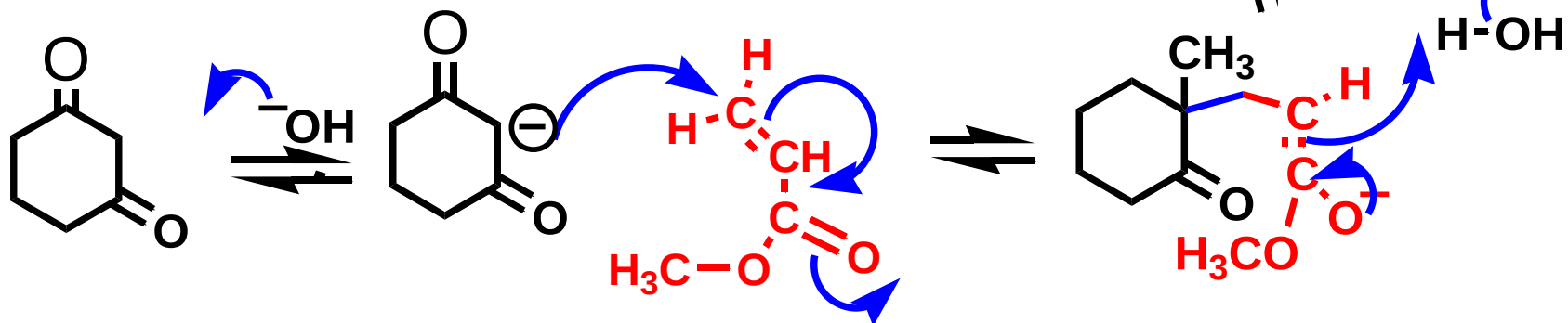




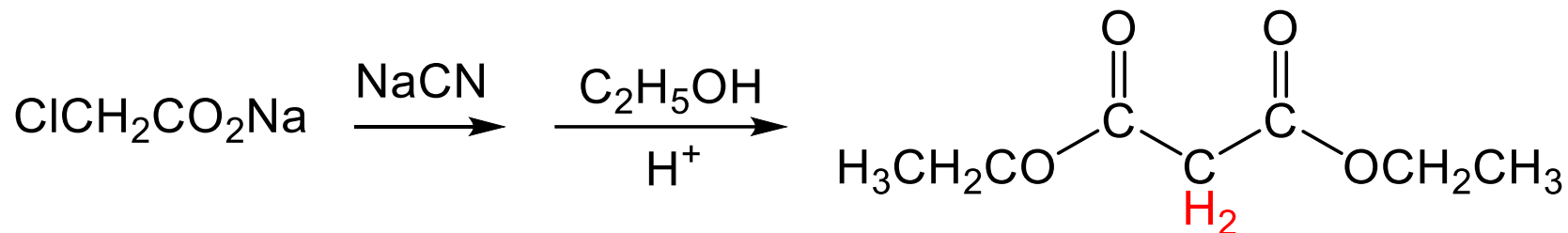
Michael 加成反应



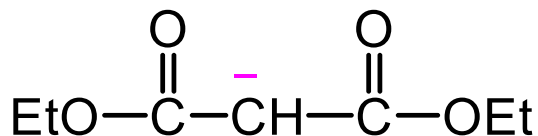
Mechanism:



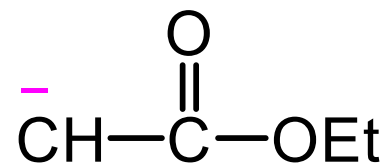
丙二酸酯以及酮式分解的应用



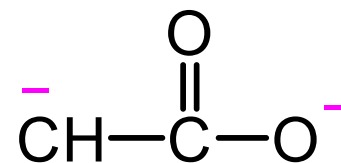
合成取代乙酸

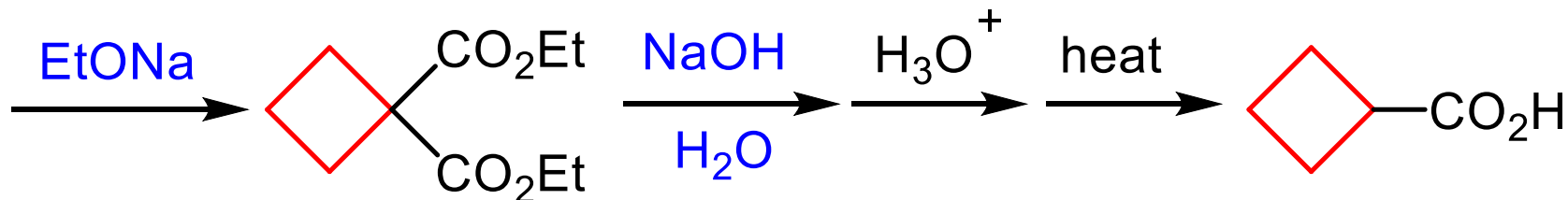
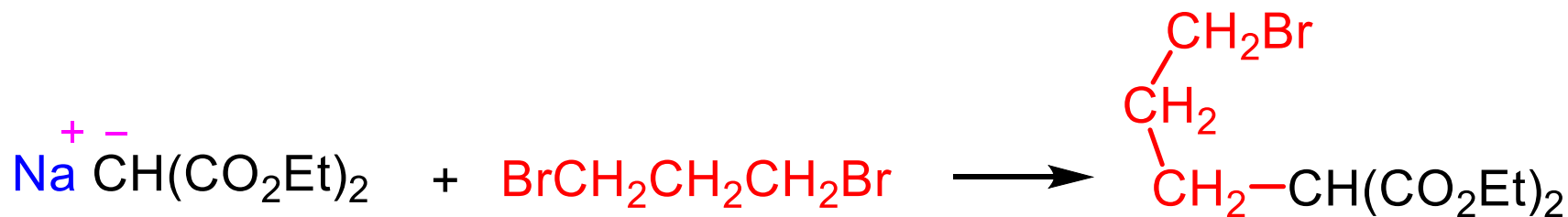
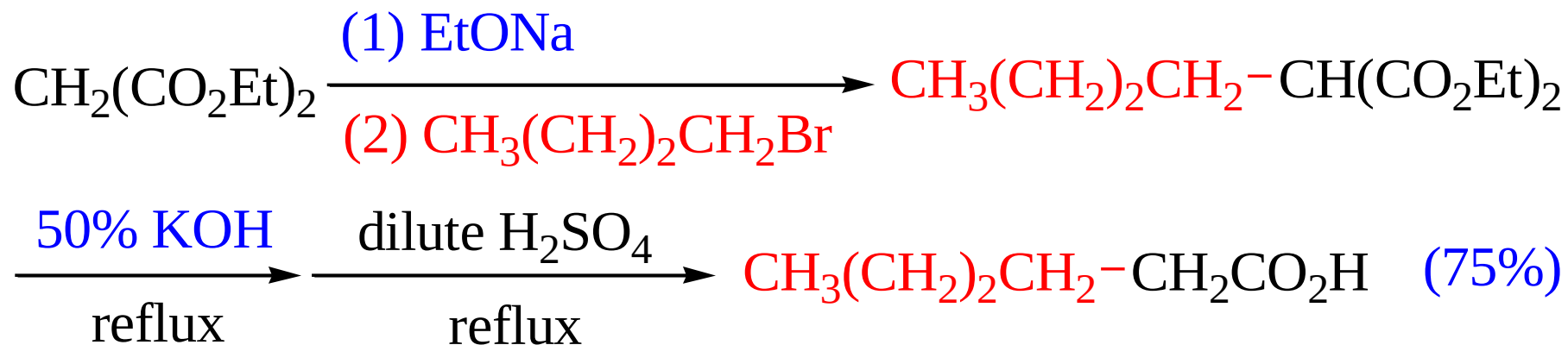


is the synthetic equivalent of

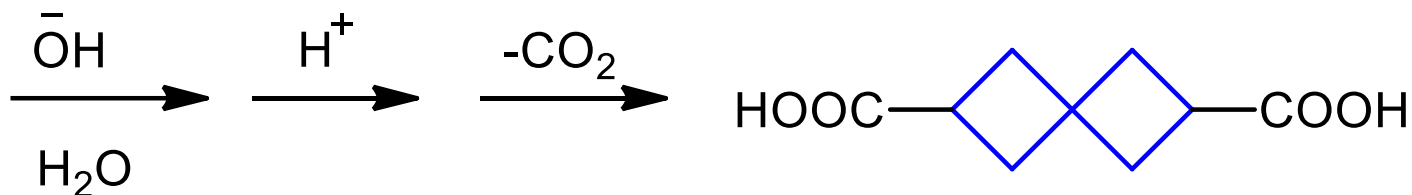
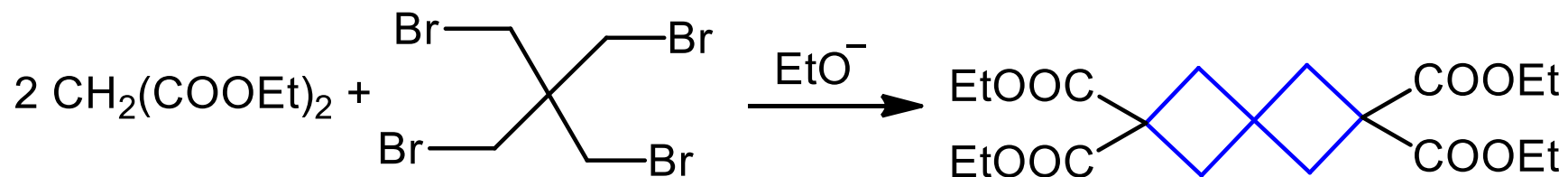


and

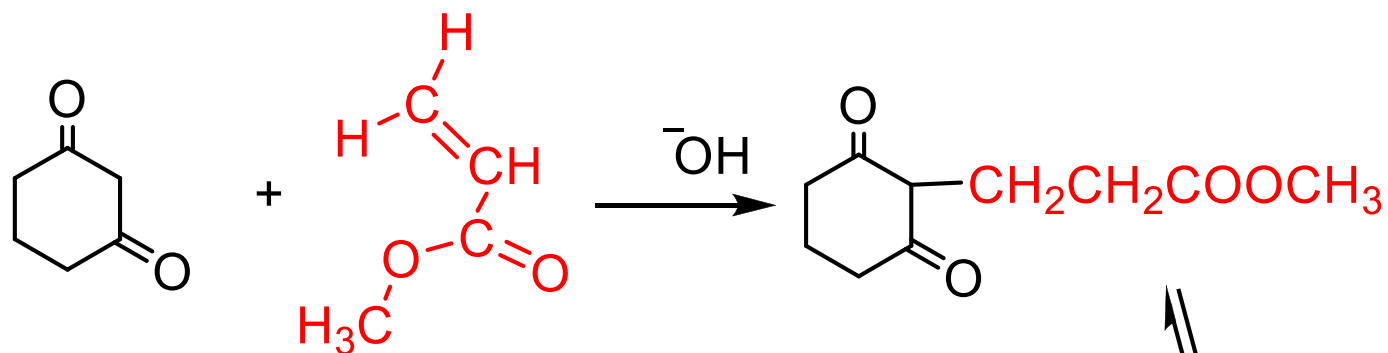




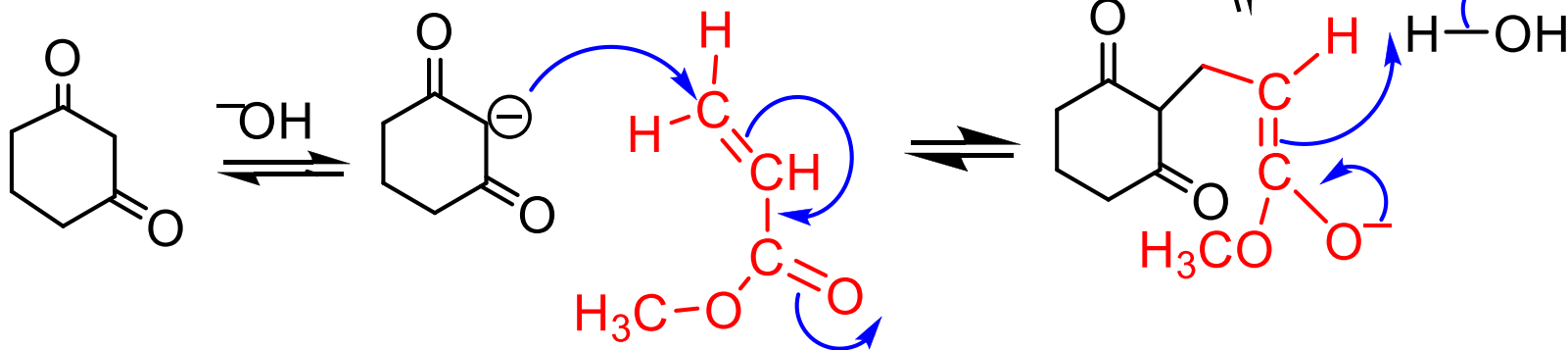
由丙二酸二乙酯为起始原料合成螺[3.3]庚烷-2,6-二羧酸：



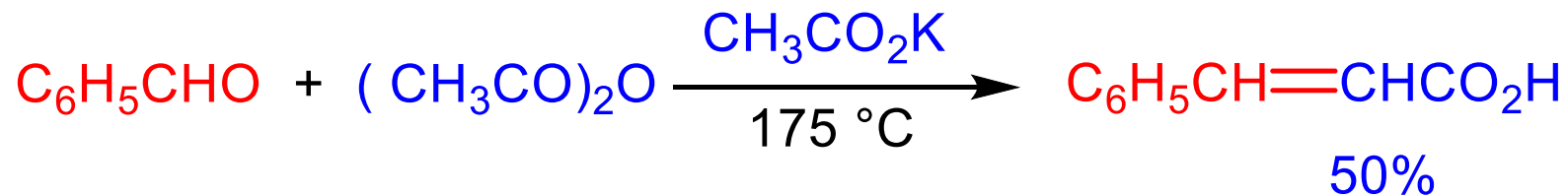
Micheal 加成反应



Mechanism:

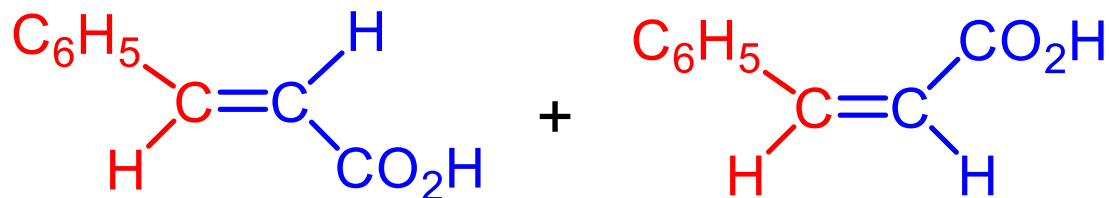


Perkin 反应



Cinnamic acid (肉桂酸)

Perkin reaction



Major

Minor

反应机理

